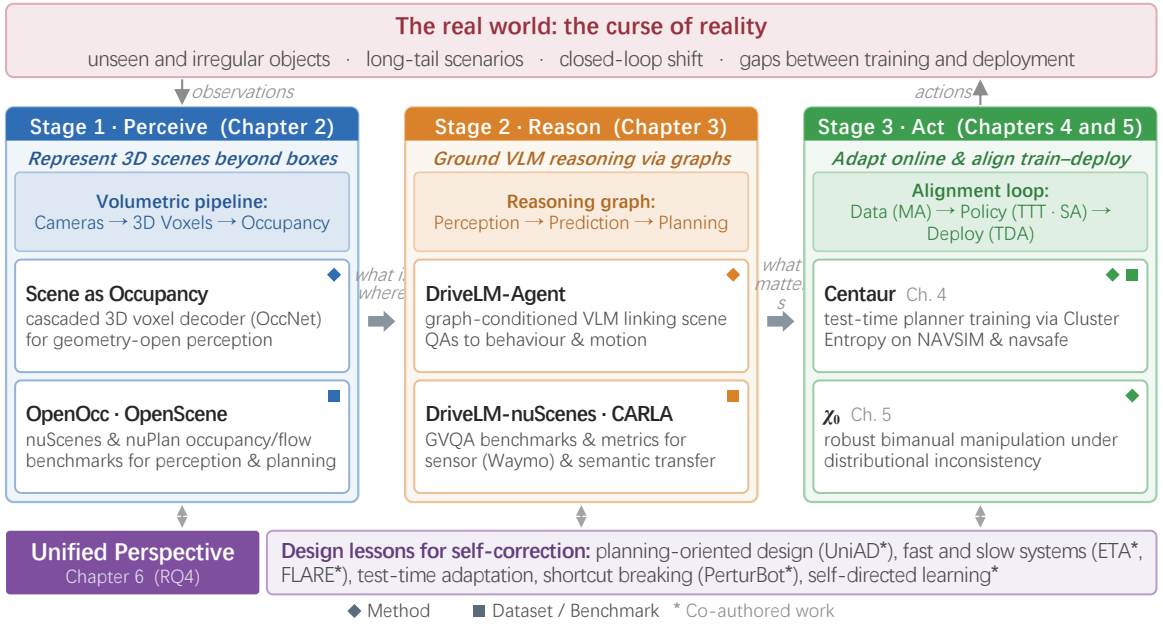

The real world: the curse of reality
unseen and irregular objects · long-tail scenarios · closed-loop shift · gaps between training and deployment
observations
actions
Stage 1 · Perceive (Chapter 2)
Represent 3D scenes beyond boxes
Volumetric pipeline:
Cameras → 3D Voxels → Occupancy
Scene as Occupancy
cascaded 3D voxel decoder (OccNet) for geometry-open perception
OpenOcc · OpenScene
nuScenes & nuPlan occupancy/flow benchmarks for perception & planning
what is where
Stage 2 · Reason (Chapter 3)
Ground VLM reasoning via graphs
Reasoning graph:
Perception → Prediction → Planning
DriveLM-Agent
graph-conditioned VLM linking scene QAs to behaviour & motion
DriveLM-nuScenes · CARLA
GVQA benchmarks & metrics for sensor (Waymo) & semantic transfer
what matters
Stage 3 · Act (Chapters 4 and 5)
Adapt online & align train–deploy
Alignment loop:
Data (MA) → Policy (TTT · SA) → Deploy (TDA)
Centaur Ch. 4
test-time planner training via Cluster Entropy on NAVSIM & navsafe
χ0 Ch. 5
robust bimanual manipulation under distributional inconsistency
Unified Perspective
Chapter 6 (RQ4)
Design lessons for self-correction: planning-oriented design (UniAD*), fast and slow systems (ETA*, FLARE*), test-time adaptation, shortcut breaking (PerturBot*), self-directed learning*
◆ Method ■ Dataset / Benchmark * Co-authored work